

BACSE291 - INNOVATIVE DESIGN PROJECT

COMMON GUIDELINES, REVIEW SCHEDULE AND EVALUATION RUBRICS

ACADEMIC YEAR 2026-2027 B.Tech CSE and its Specializations

1. Course Context

BACSE291 - Innovative Design Project is a project-based course for second-year B.Tech. students. In accordance with the IDP circular, it focuses on problem-solving, design thinking and the development of a functional prototype with moderate technical complexity. The prescribed mode of evaluation is Reviews and Report Writing.

Course Objectives

- 1: Apply engineering design principles to real-world problems.
- 2: Develop engineering models and prototypes using appropriate tools.
- 3: Conduct testing, validation and optimisation of designs.
- 4: Prepare technical reports and deliver professional presentations.

Course Outcomes

- CO1: Apply engineering design methodology to solve practical problems by integrating creativity with technical feasibility.
- CO2: Demonstrate skills in modelling, simulation, prototyping and testing, and evaluate solutions for performance, safety, cost-effectiveness and sustainability.
- CO3: Demonstrate teamwork and project management skills.
- CO4: Communicate engineering solutions effectively through reports and presentations.

Project Framework

- Assessment is conducted through the prescribed seven review schedule, with a total of 100 marks across the academic year.
- Each project team shall consist of 2 to 3 students, in accordance with the approved syllabus.
- The project shall address a real-world engineering problem and develop an innovative, feasible and sustainable solution through systematic design and implementation. The final outcome may be a functional prototype, software application, process model or design solution, supported by testing, validation and documentation.
- Each team shall demonstrate progress in every prescribed review and submit the final project report in the format circulated by the coordinators.

2. Common Project Guidelines

1. Each project team shall define a realistic problem statement, scope, objectives, expected outcomes and work plan for an innovative, feasible and sustainable solution.
2. Each project group shall consist of two to three students under the guidance of an assigned faculty mentor. Group formation shall follow the School/Programme arrangements approved by the Dean.
3. Students shall meet their assigned faculty mentor regularly, maintain progress records and use the feedback received to improve the project work.
4. All team members shall actively participate and contribute equitably throughout the project duration, including project work, reviews, demonstrations and report preparation. Project evaluation shall primarily reflect the quality of the overall team deliverable. Where applicable, individual performance may be differentiated based on demonstrated contribution, technical understanding and response during the review.
5. Each team shall follow the prescribed review schedule, demonstrate the expected progress at every stage, address review feedback, and submit the final report in the prescribed format.
6. Model development at **Technology Readiness Level (TRL) 3** is encouraged. **Publication in a journal or conference is not mandatory.** Projects demonstrating significant novelty may be considered for patent filing. The standard project-report format will be circulated by the coordinators.

3. Review Schedule and Marks Distribution

Semester	Review	Evaluation Period	Evaluator	Marks	Expected Outcome
Fall 2026-27	Review I	17-21 Aug 2026	Guide	5	Identify a real-world problem; justify the need; define objectives, scope and expected outcomes; present preliminary literature survey and methodology.
Fall 2026-27	Review II	21-25 Sep 2026	Panel constituted by the School	20	Complete approximately 20% of the project, including requirement analysis, system design, component selection and initial prototype/module development.
Fall 2026-27	Review III	12-16 Oct 2026	Guide	10	Complete approximately 30% of the project, including refined requirement analysis, system design, justified component/tool selection and initial prototype/module development; demonstrate follow-up on Review II observations.
Winter 2026-27	Review IV	25-29 Jan 2027	Guide	15	Complete approximately 50% of the project, with major modules implemented and integrated to demonstrate core functionality.
Winter 2026-27	Review V	8-12 Mar 2027	Panel constituted by the School	25	Complete approximately 80% of the project; integrate planned modules; validate functionality through testing; demonstrate a complete working prototype.
Winter 2026-27	Review VI	29 Mar-2 Apr 2027	External Panel (Open House)	15	Present a fully functional prototype during the Open House for independent external evaluation, incorporating prior feedback and highlighting technical quality, performance evaluation, innovation, societal or industrial relevance and future enhancements.
Winter 2026-27	Report Submission	2 Apr 2027	Guide	10	Submit the comprehensive final report, including problem definition, literature review, methodology, implementation, testing, results, innovation, conclusions, future scope and references.
Overall Total				100	

4. Evaluation Rubrics - FALL Semester 2026-2027

Review I - Problem Identification and Planning - 5 Marks

Evaluation Parameter	Marks	Assessment Focus	Mapped CO(s)
Problem relevance and justification	1	Real-world need and significance of the proposed solution.	CO1
Objectives, scope and expected outcomes	1	Clear, achievable and appropriately scoped project plan.	CO1
Preliminary literature survey	1	Relevant study of existing solutions and limitations.	CO1
Proposed methodology and feasibility	1	Suitable approach, preliminary design direction and feasibility.	CO1, CO2
Work plan and team responsibilities	1	Realistic timeline and clear allocation of roles.	CO3
Total	5		

Review II - Initial Design and Development - 20 Marks

Evaluation Parameter	Marks	Assessment Focus	Mapped CO(s)
Requirement analysis and problem understanding	3	Clarity of the problem, completeness of requirements, user/stakeholder needs and relevant constraints.	CO1

Evaluation Parameter	Marks	Assessment Focus	Mapped CO(s)
System design and architecture	3	System architecture, module definition, interfaces and suitability of the proposed technical design.	CO1, CO2
Component/tool selection and technical justification	3	Appropriate selection of hardware, software, datasets, methods or tools, supported by sound technical reasons.	CO2
Initial prototype or module development	3	Evidence of approximately 20% implementation through demonstrable modules, prototype elements or proof of concept.	CO2
Innovation and feasibility	3	Originality, practicality, resource feasibility and potential value of the proposed solution.	CO1, CO2
Project planning, teamwork and presentation	3	Realistic work plan, clear team responsibilities, coordinated participation and coherent presentation of progress.	CO3, CO4
Individual contribution and technical response	2	Individual evidence of assigned work, technical understanding and ability to respond to questions.	CO3, CO4
Total	20		

Review III - Progress Review - 10 Marks

Evaluation Parameter	Marks	Assessment Focus	Mapped CO(s)
Follow-up on Review II and progress towards 30% completion	2	Evidence that Review II observations have been addressed and measurable 30% completion has been achieved through updated plans, completed tasks, artefacts or demonstrations.	CO2, CO3
Requirement analysis and system design refinement	2	Updated requirements and a refined system design/architecture aligned with the approved problem, scope and constraints.	CO1, CO2
Component/tool selection and technical justification	2	Appropriate confirmation or refinement of selected hardware, software, datasets, methods or tools, supported by sound technical reasons.	CO2
Initial prototype/module development and early testing	2	Demonstrable initial prototype or module development, with early testing, identified challenges and suitable corrective actions.	CO2
Individual contribution, technical understanding and next-stage responsibility	2	Individual evidence of completed assigned work and contribution to the 30% milestone, technical understanding of the work, and clear responsibility for next-stage tasks.	CO3, CO4
Total	10		

5. Evaluation Rubrics - Winter Semester 2026-2027

Review IV - Core Functionality and Integration - 15 Marks

Evaluation Parameter	Marks	Assessment Focus	Mapped CO(s)
Progress towards 50% completion	3	Major planned tasks and modules completed in accordance with the approved work plan.	CO2, CO3
Core functionality	3	Demonstrable working functions that address the defined problem and project objectives.	CO1, CO2

Evaluation Parameter	Marks	Assessment Focus	Mapped CO(s)
Module integration and interface coherence	3	Effective integration of implemented modules/components with coherent data, control or user interaction flows.	CO2
Technical quality and refinement	2	Proper implementation, appropriate technical decisions and refinement based on development evidence.	CO1, CO2
Documentation, teamwork and next-stage readiness	2	Current documentation, coordinated team progress and a realistic plan for the next milestone.	CO3, CO4
Individual contribution and technical understanding	2	Individual evidence of assigned work completed for the 50% milestone, technical understanding and readiness for assigned next-stage tasks.	CO3, CO4
Total	15		

Review V - Integrated Prototype and Validation - 25 Marks

Evaluation Parameter	Marks	Assessment Focus	Mapped CO(s)
Progress towards 80% completion and integration	5	Substantial completion and integration of the planned system, prototype or solution.	CO2, CO3
Prototype functionality and completeness	5	Demonstrable end-to-end functions, feature completeness and alignment with stated requirements.	CO1, CO2
Performance, reliability and usability	5	Evidence of solution performance, stability, usability or other relevant quality attributes.	CO2
Testing and validation evidence	4	Appropriate test strategy, test cases, results, validation evidence and interpretation.	CO2
Innovation, feasibility, relevance and team presentation	4	Innovation, practical feasibility, real-world relevance and coherent team presentation of the integrated solution.	CO1, CO2, CO4
Individual contribution and technical response	2	Individual evidence of assigned contribution to the integrated prototype and ability to respond to technical questions.	CO3, CO4
Total	25		

Review VI - Open House Evaluation - 15 Marks

Evaluation Parameter	Marks	Assessment Focus	Mapped CO(s)
Functional prototype and end-to-end demonstration	3	A complete working prototype, application, process model or design solution demonstrated independently.	CO2
Technical quality, integration and refinement	3	Sound implementation, coherent integration, appropriate technical choices and visible refinement based on earlier feedback.	CO1, CO2
Innovation, relevance and practical value	3	Originality, societal or industrial relevance, feasibility and value of the proposed solution.	CO2
Testing, performance and validation evidence	2	Credible evidence of testing, functionality, performance and validation of the completed solution.	CO1, CO2
Communication, demonstration readiness and future scope	2	Clear team demonstration, effective communication and a realistic future direction.	CO3, CO4
Individual technical ownership and response	2	Individual technical ownership and response to questions demonstrated during the external-panel Open House evaluation.	CO3, CO4

Evaluation Parameter	Marks	Assessment Focus	Mapped CO(s)
Total	15		

Report Submission - 10 Marks

Evaluation Parameter	Marks	Assessment Focus	Mapped CO(s)
Problem definition and literature review	2	Clear context, need, objectives and relevant prior work.	CO1
Design methodology and implementation	2	Sound technical approach, design rationale and implementation details.	CO1, CO2
Testing, results and analysis	2	Evidence, results, validation and appropriate interpretation.	CO2
Innovation, conclusions and future scope	2	Project contribution, justified conclusions, practical relevance and realistic future work.	CO1, CO2
Organisation, format, references and technical presentation	2	Compliance with the prescribed report format, logical organisation, technical clarity and proper referencing.	CO4
Total	10		